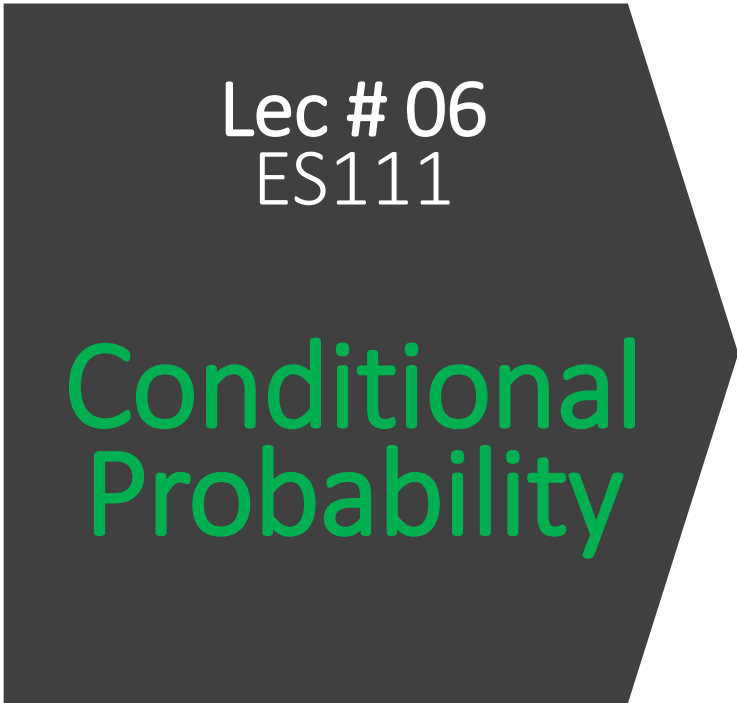




Lec # 06
ES111

Conditional
Probability

@M. Tahir Naseem



Discussion Plan

1. **Conditional** Probability

1. **Independent** Events

3. **Product** Rule

(Sec. 2.6)



Probability: Assigning weights to sample points

1. Probability of event A , sum of all weights of sample points in A

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$$S = \{1, 2, 3, 4, 5, 6\}$$

(i) assign weight to sample points

(ii) sum of all weights must be 1

(iii) Identify E , and sum all weights of E

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$$S = \{ \underset{\substack{\downarrow \\ w}}{1}, \underset{\substack{\downarrow \\ w}}{2}, \underset{\substack{\downarrow \\ w}}{3}, 4, 5, 6 \}$$

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$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $w \quad 2w \quad w \quad 2w \quad w \quad 2w$

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$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $w \quad 2w \quad w \quad 2w \quad w \quad 2w$

(i) assign weight to sample points $\rightarrow 9w$

(ii) sum of all weights must be 1

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(i) assign weight to sample points $\rightarrow 9w$

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(i) assign weight to sample points $\rightarrow 9w$

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(i) assign weight to sample points $\rightarrow 9w$

(ii) sum of all weights must be 1 $\rightarrow 9w = 1 \rightarrow w = \frac{1}{9}$

(iii) Identify E , and sum all weights of E $P(E) = \frac{1}{9} + 2w + \frac{1}{9} + 2w$

Probability: Assigning weights to sample points

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(iii) Identify E , and sum all weights of E $P(E) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9}$

Ans

Probability for equally likely events

$$\text{Probability of } A = \frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$$

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$$P(A) =$$

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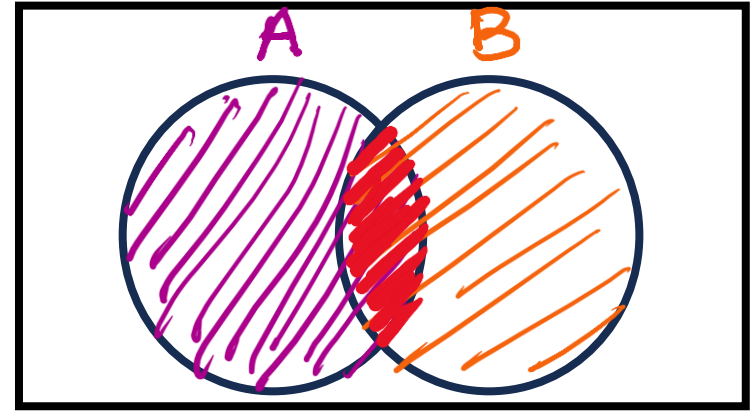
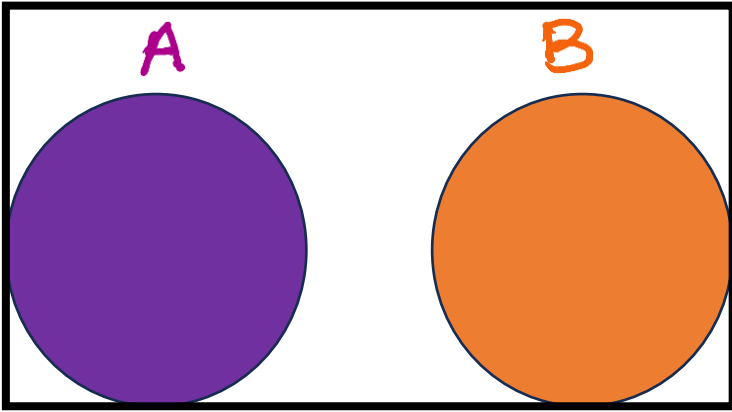
$$P(A) = \frac{n}{N}$$

Additive Rules of Probability

Let A and B are two events, we want to add the probabilities

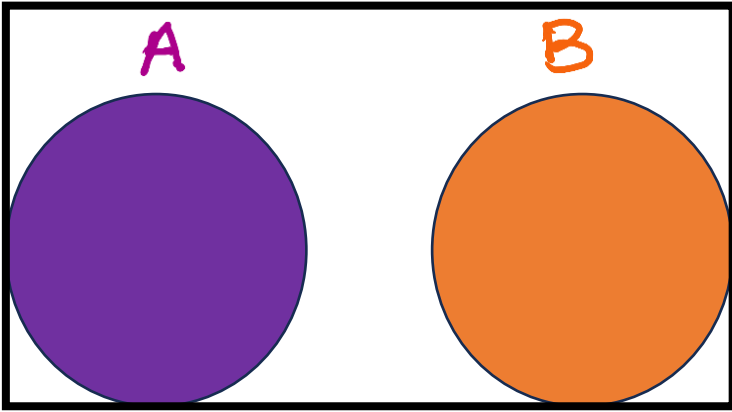
Additive Rules of Probability

$A \cup B$

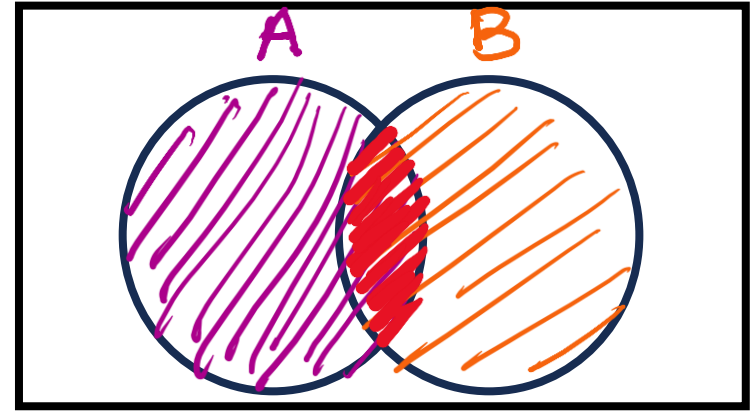


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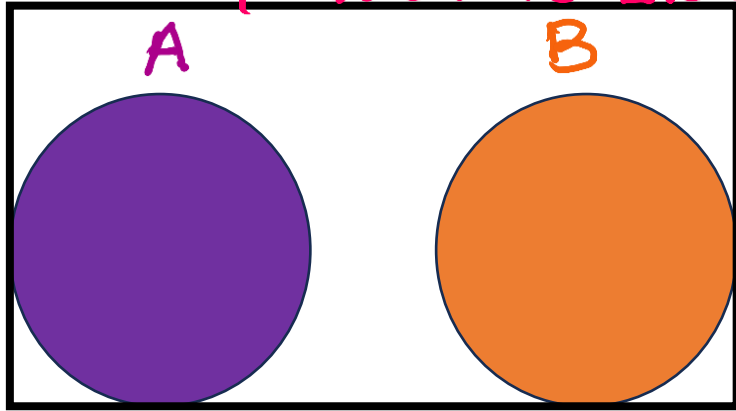


$$A \cap B = \{\emptyset\}$$

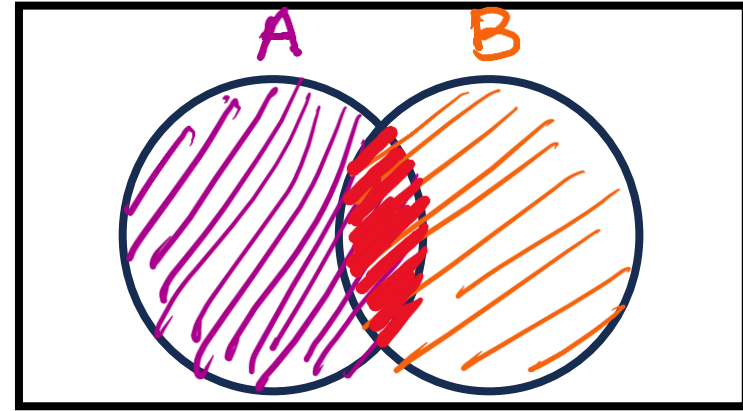


Additive Rules of Probability

$A \cup B$ Mutually Exclusive Events

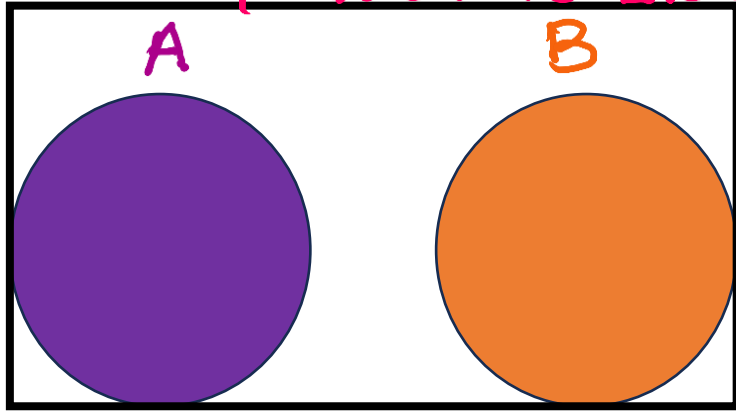


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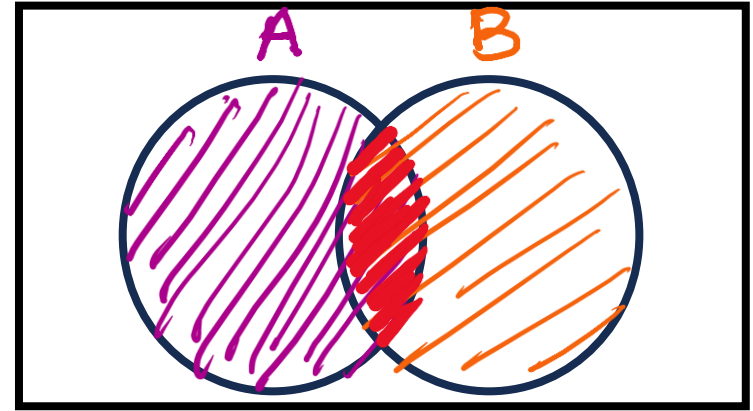


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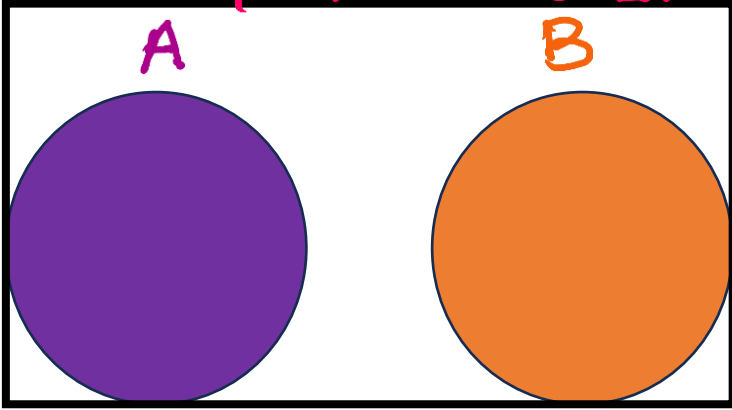
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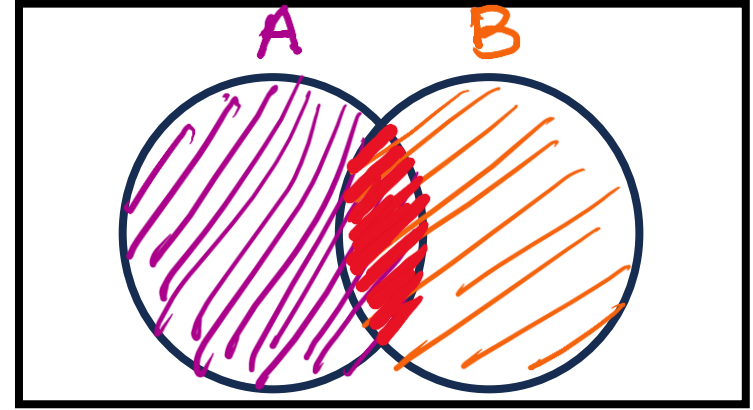
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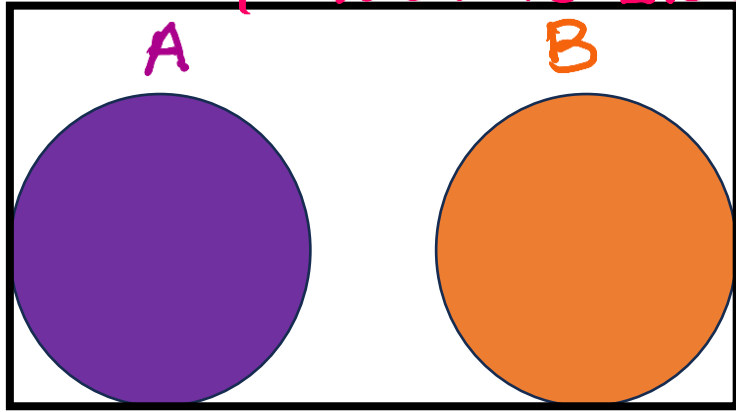


$$A \cap B \neq \{\emptyset\}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

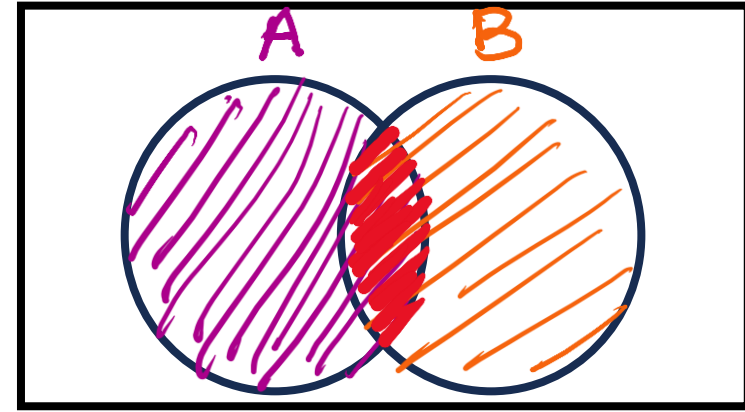
Additive Rules of Probability

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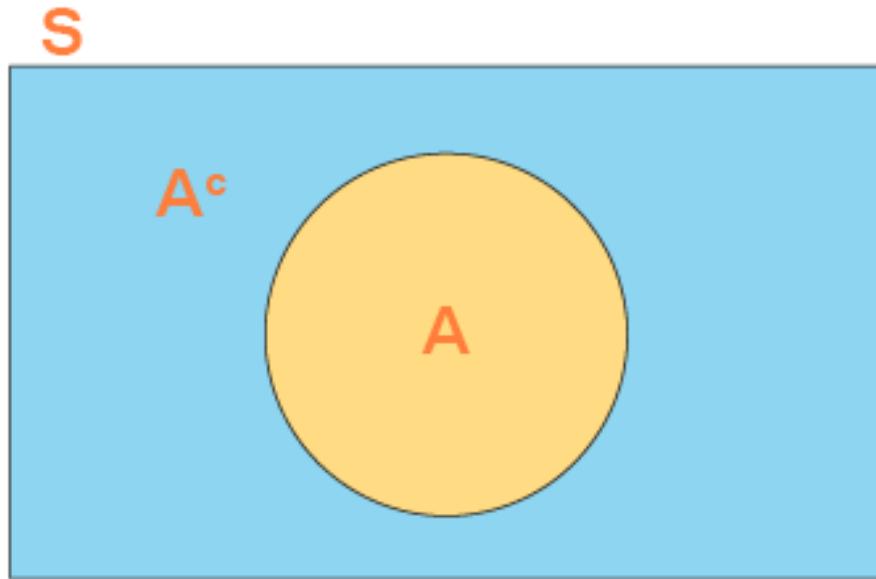


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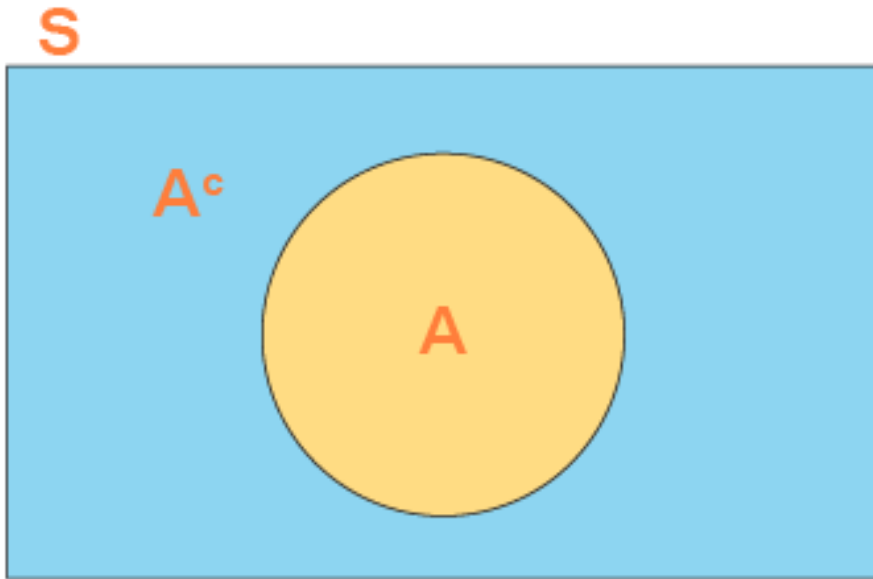
Additive Rules of Probability

If A and A' are complementary events



Additive Rules of Probability

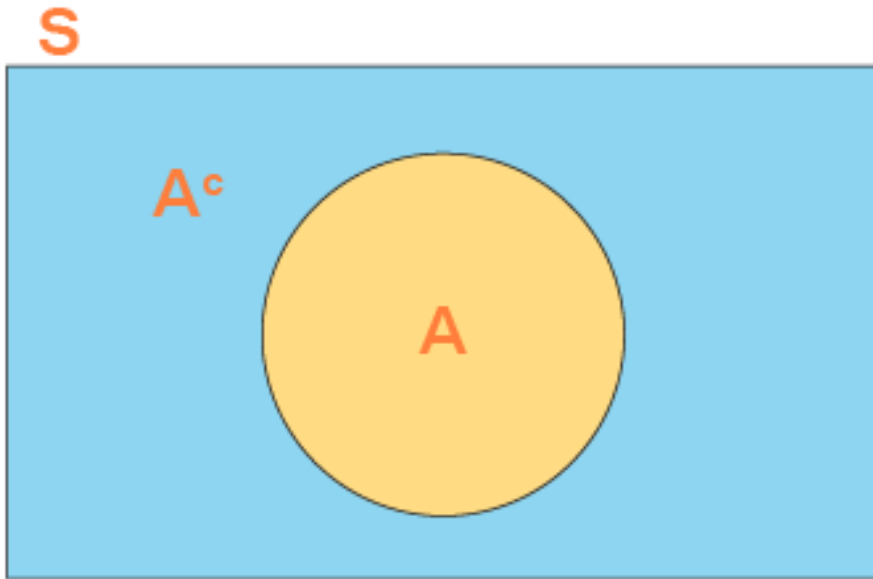
If A and A' are complementary events



$$P(A) + P(A') =$$

Additive Rules of Probability

If A and A' are complementary events



$$P(A) + P(A') = P(S)$$

conditional Probability

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conditional Probability

Q: What is the probability to caught
covid virus in swabi ?



conditional Probability

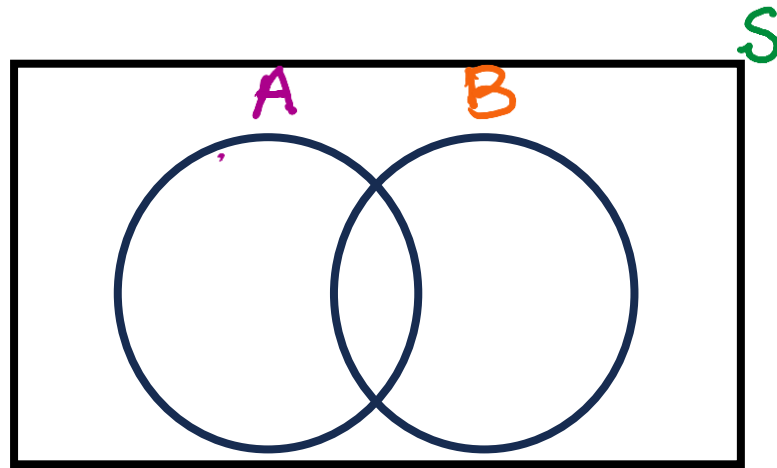
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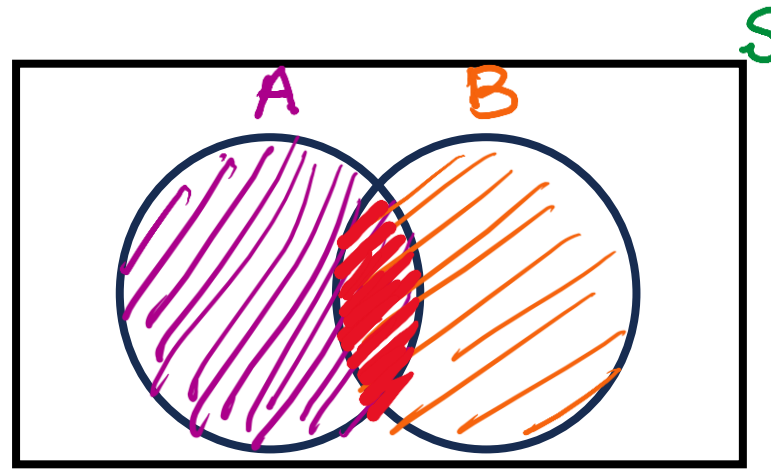
Q: What is the probability to caught
covid virus in swabi, given the age of person less than five years ?

conditional Probability

conditional Probability



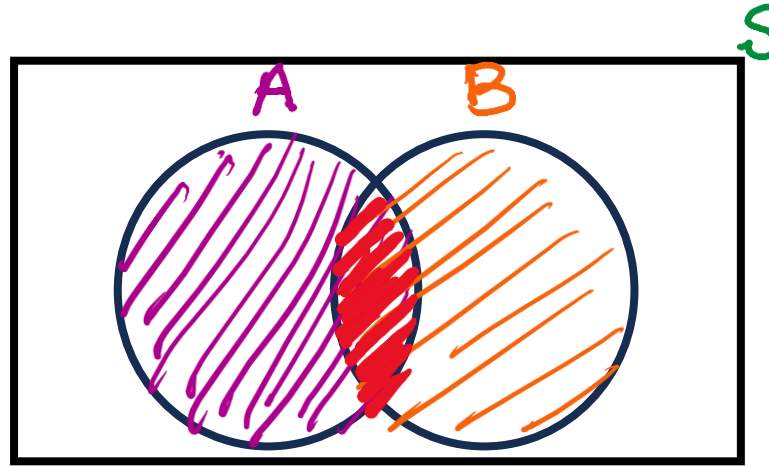
conditional Probability



we want to find

$$P(A/B) =$$

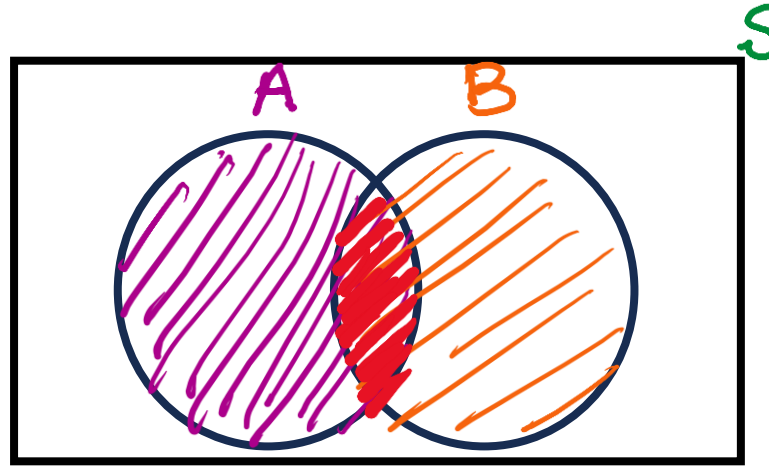
conditional Probability



we want to find

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

conditional Probability



we want to find

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \rightarrow \text{why?}$$



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$$P(E) = \frac{2}{9}$$

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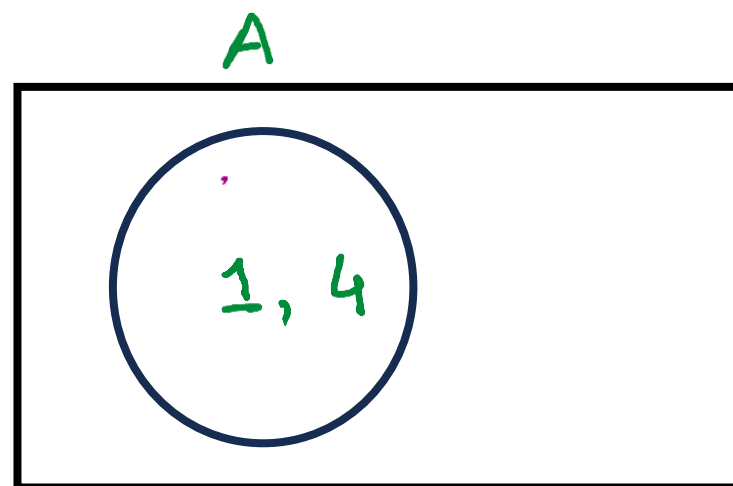
without "condition"

$$P(A) = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}$$

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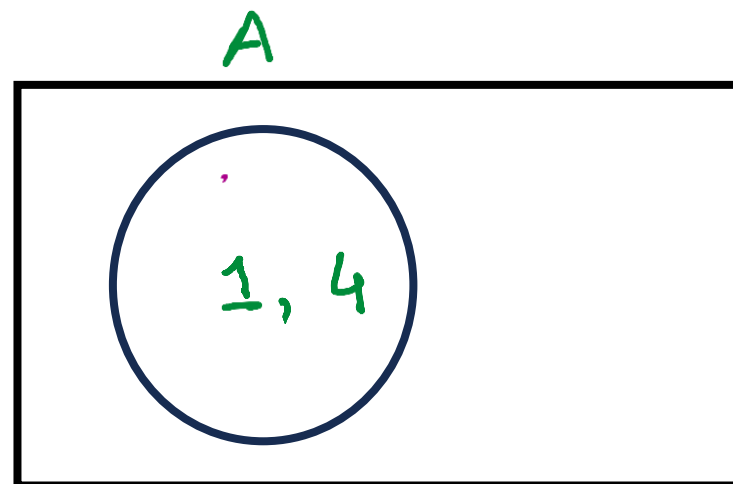
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Suppos now "B" has occurred

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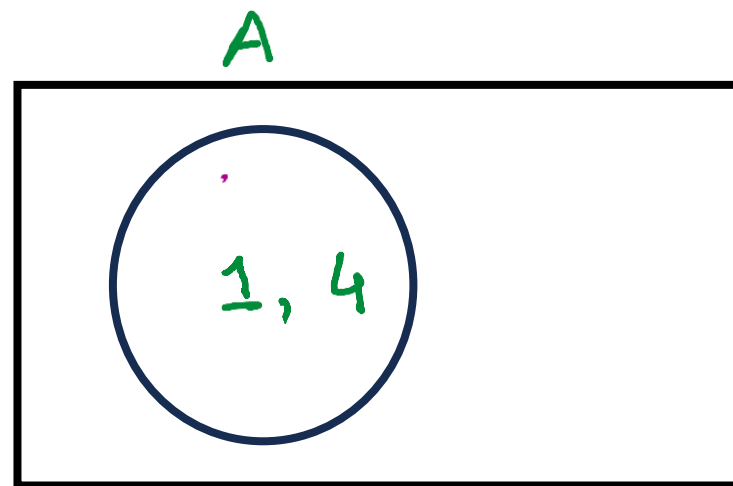
Suppos now "B" has occurred

$$B = \{4, 5, 6\}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

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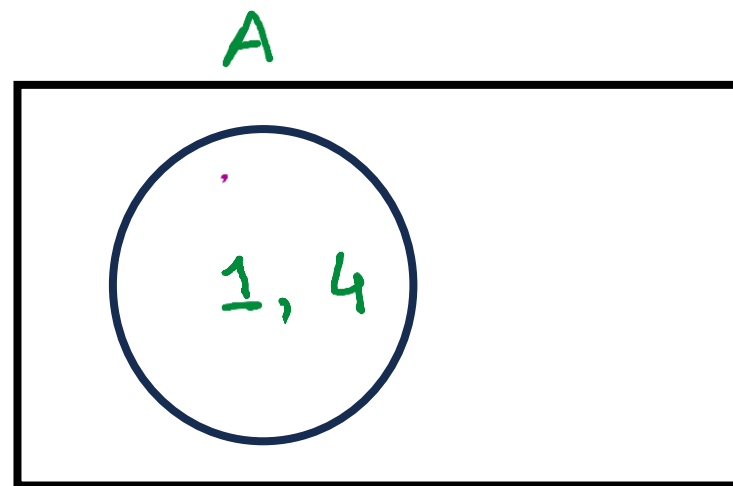
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$B = \{4, 5, 6\}$ → This will be "new sample space"

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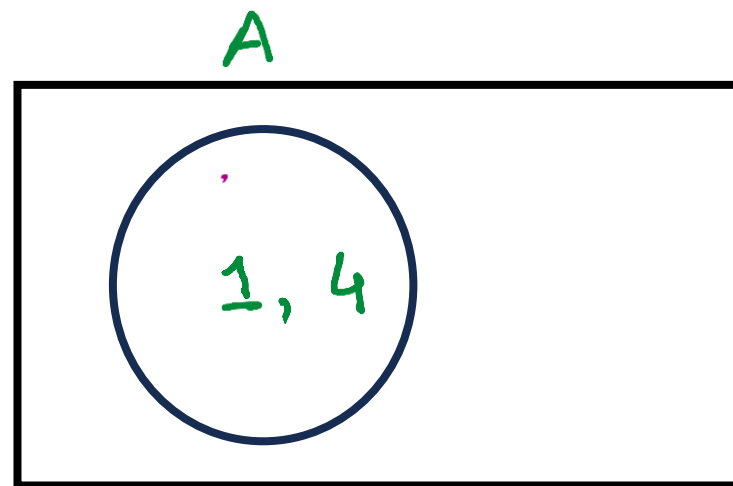
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↙ ↘
2w 2w

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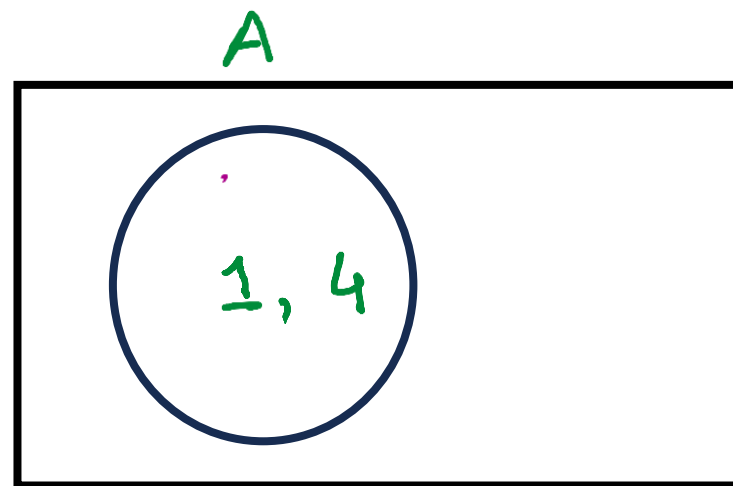
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2W W 2W

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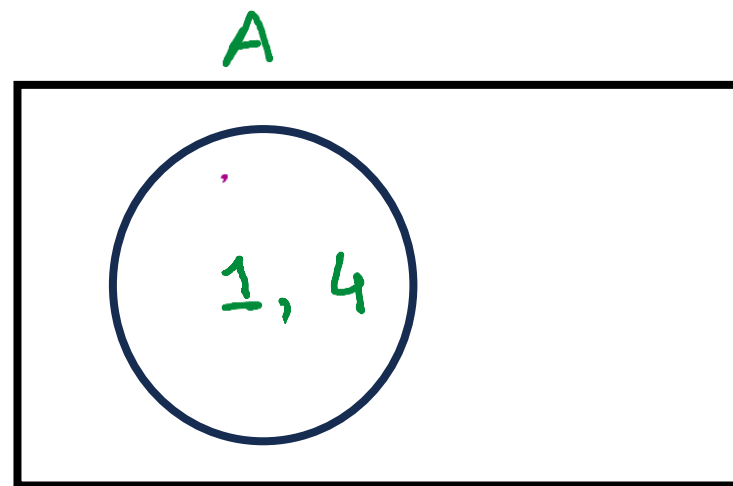
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↙ ↘ ↙ ↘
2W W 2W

$$5W = 1$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

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↙ ↘ ↙ ↘
2W W 2W

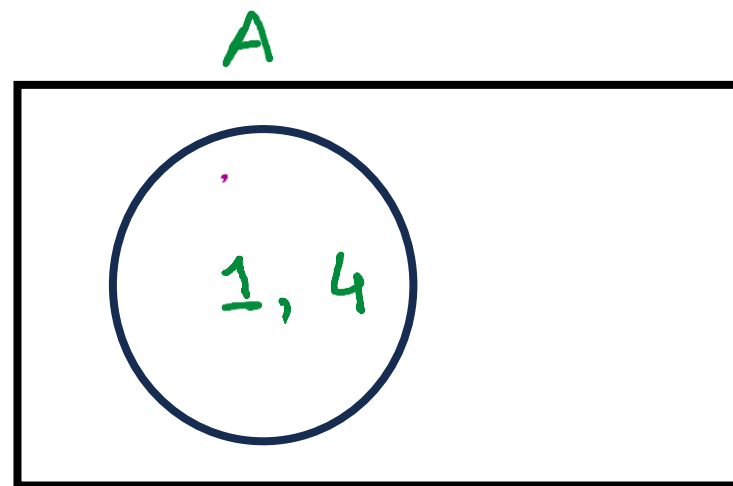
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$$S = \{1, 2, 3, 4, 5, 6\}$$

Without "condition"

$$P(A) = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}$$

Suppose now "B" has occurred

$$B = \{4, 5, 6\} \rightarrow \text{This will be "new sample space"}$$

← 2W
← W
← 2W

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↙ ↘ ↙
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$$P(A/B) = \frac{2}{5} \quad \underline{\underline{\text{Ans}}}$$

$$A = \{1, 4\}$$



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$$P(A/B) = \frac{2}{5} \quad \underline{\underline{\text{Ans}}}$$

$$A = \{1, 4\}$$

$$B = \{4, 5, 6\}$$



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$B = \{4, 5, 6\}$ → This will be "new sample space"
↙ ↘ ↙
2W W 2W

$$5W = 1$$

$$W = \frac{1}{5}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(A/B) = \frac{2}{5} \quad \underline{\underline{\text{Ans}}}$$

$$A = \{1, 4\}$$

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A die is loaded in such a way that an **even number** is twice as likely to occur **as an odd number**. **What is the probability of getting perfect square given the die resulted in a number greater than 3.**

Without "condition"

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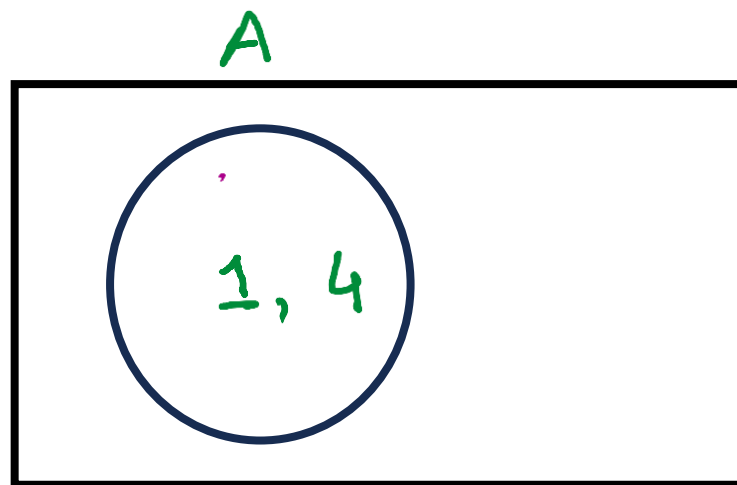
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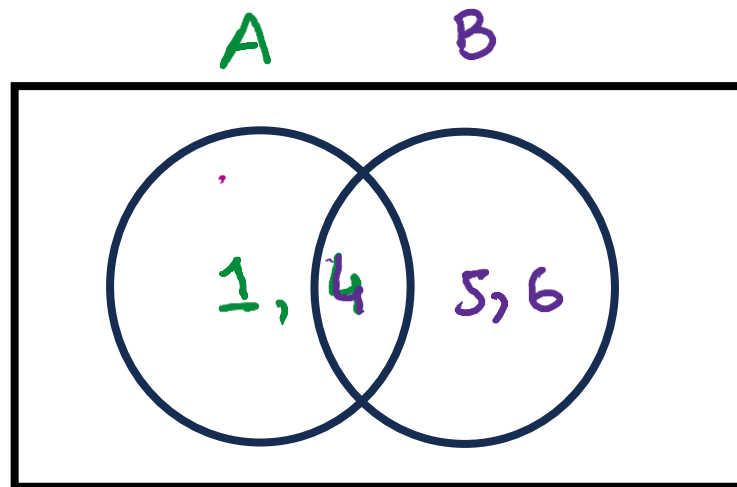
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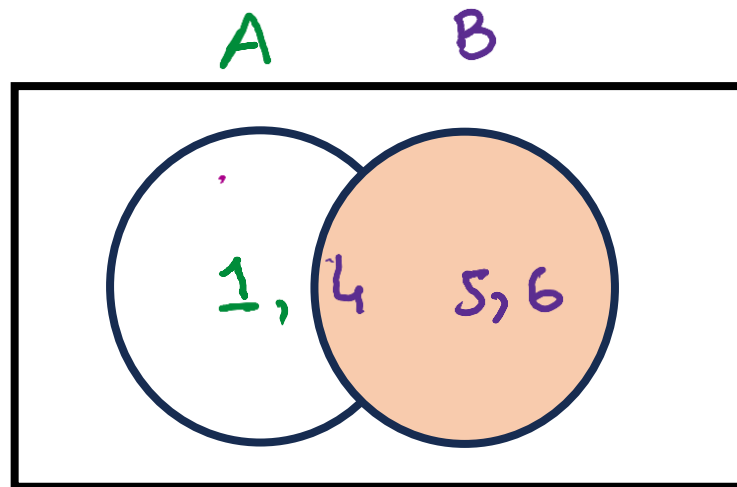
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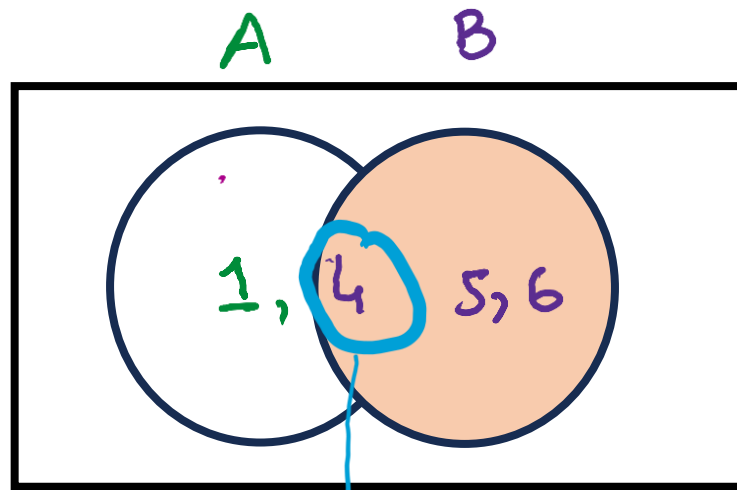
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The probability that a regularly scheduled flight departs on time is $P(D) = 0.83$; the probability that it arrives on time is $P(A) = 0.82$; and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$. Find the probability that a plane (a) arrives on time, given that it departed on time, and (b) departed on time, given that it has arrived on time.



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$$P(A|D) = \frac{P(A \cap D)}{P(D)}$$



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$$P(D|A) =$$



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$$P(D|A) = \frac{P(A \cap D)}{P(A)}$$



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$$P(D|A) = \frac{P(A \cap D)}{P(A)} = \frac{0.78}{0.82} \quad \text{Ans}$$



The concept of conditional probability has countless uses in both industrial and biomedical applications. Consider an industrial process in the textile industry in which strips of a particular type of cloth are being produced. These strips can be defective in two ways, length and nature of texture. For the case of the latter, the process of identification is very complicated. It is known from historical information on the process that 10% of strips fail the length test, 5% fail the texture test, and only 0.8% fail both tests. If a strip is selected randomly from the process and a quick measurement identifies it as failing the length test, what is the probability that it is texture defective?



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$$P(T) =$$



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$$P(T/L) =$$



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$$P(T|L) = \frac{P(T \cap L)}{P(L)}$$



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$$P(T|L) = \frac{P(T \cap L)}{P(L)} = \frac{0.008}{0.1} \quad \text{Ans}$$



The probability that a married man watches a certain television show is 0.4, and the probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that
(a) a married couple watches the show



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$$P(H \cap W) =$$



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$$P(H \cap W) = P(H|W)$$



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The probability that a married man watches a certain television show is 0.4, and the probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7 Find the probability that
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$$\begin{aligned} P(H \cap W) &= P(W) P(H|W) \\ &= 0.5 \times 0.7 \quad \underline{\text{Ans}} \end{aligned}$$

Independent Events

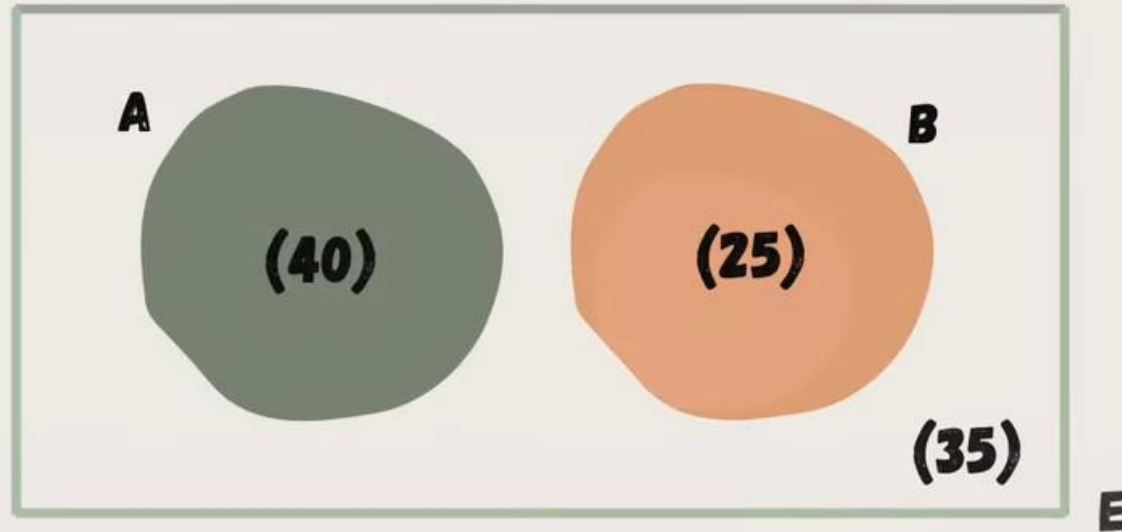
Independent Events

MUTUALLY EXCLUSIVE EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student plays
Cricket



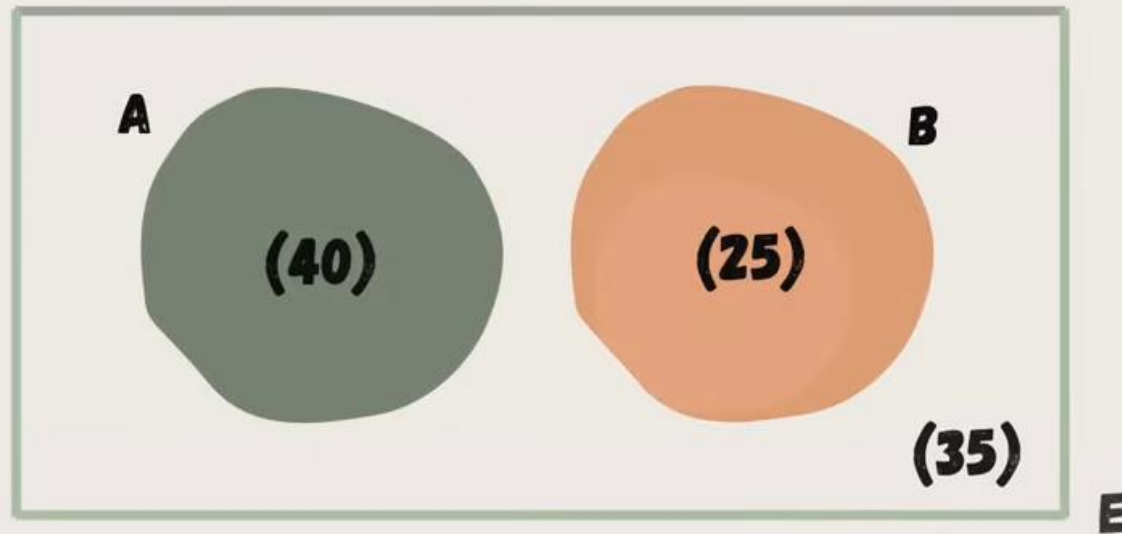
Independent Events

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$$P(A) = \frac{40}{100} = 0.4$$

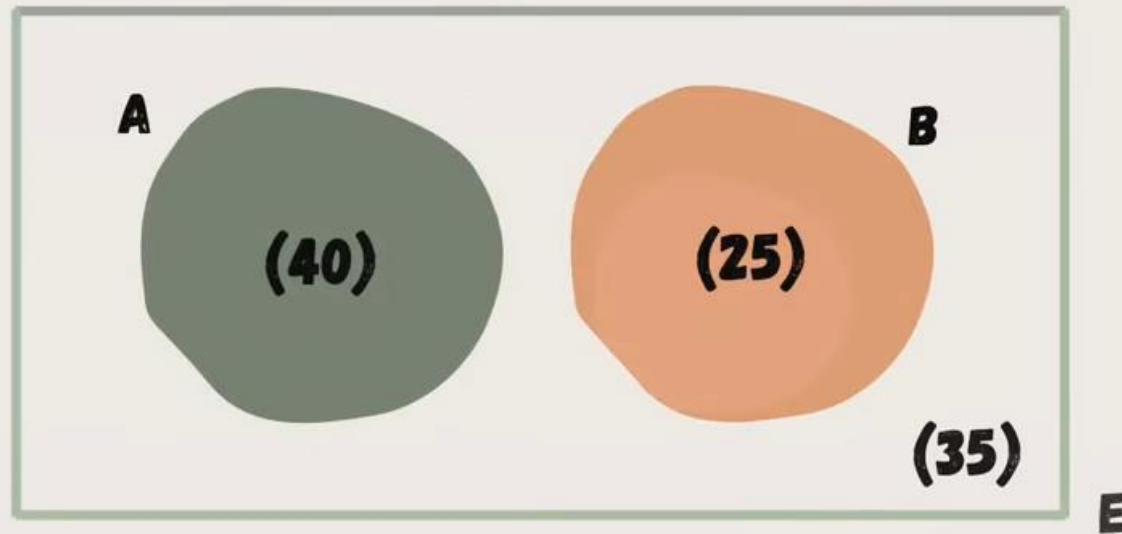
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$$P(A) = \frac{40}{100} = 0.4$$

$$P(B) = \frac{25}{100} = 0.25$$

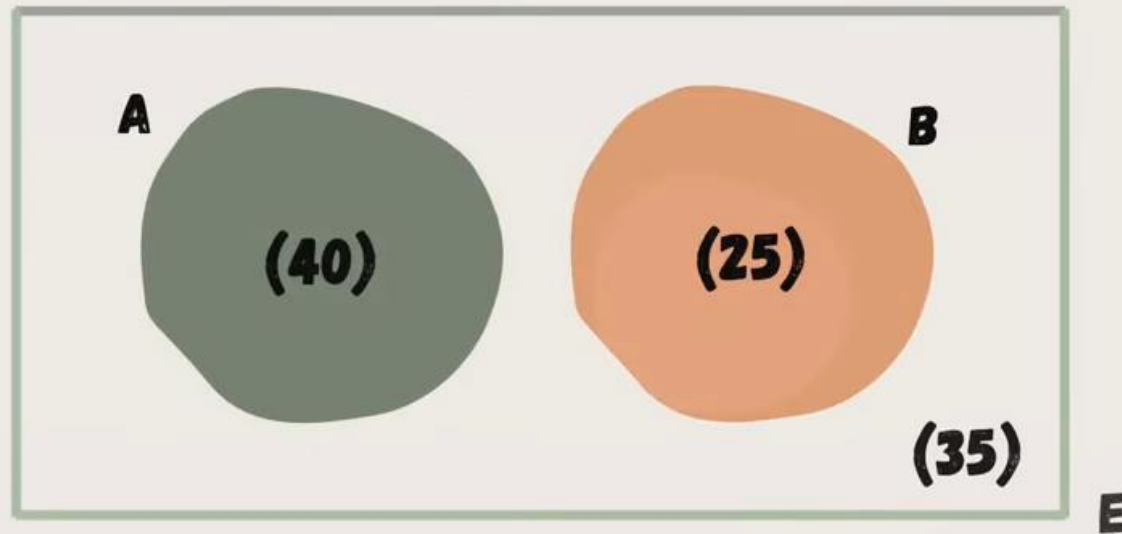
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$$P(A \cap B) = 0$$

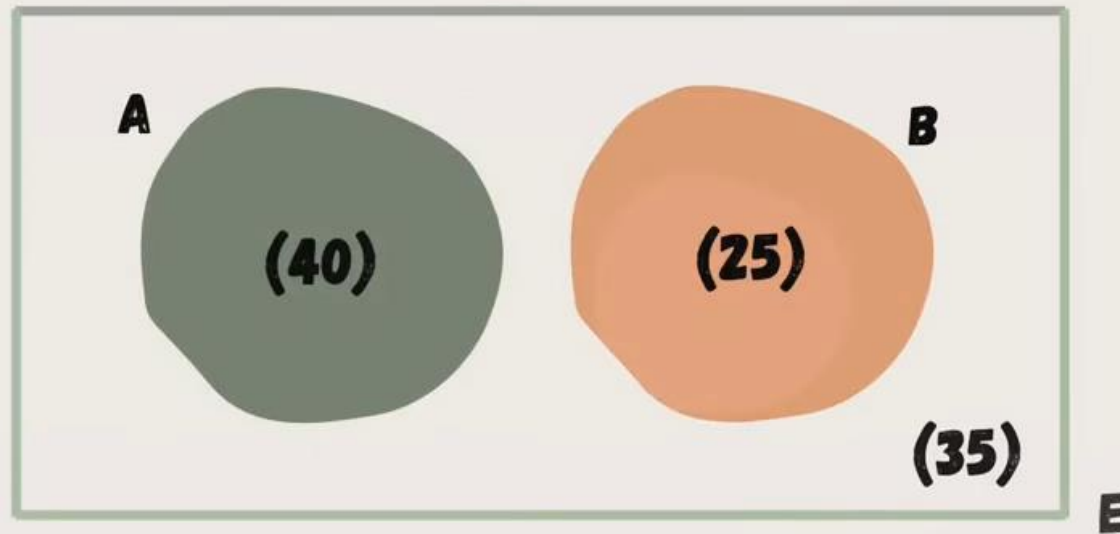
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$$P(A \cap B) = 0$$

$$P(A \cap B) = \{\emptyset\}$$

Mutually Exclusive

Independent Events

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed
→

EVENT A:
A student plays
Basketball

EVENT B:
A student studies
Modern History



Independent Events

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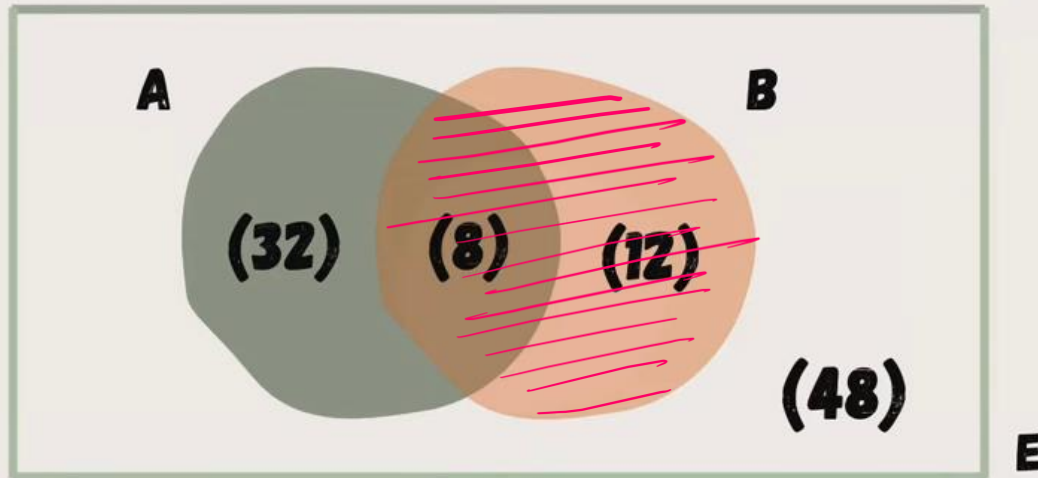
Independent Events

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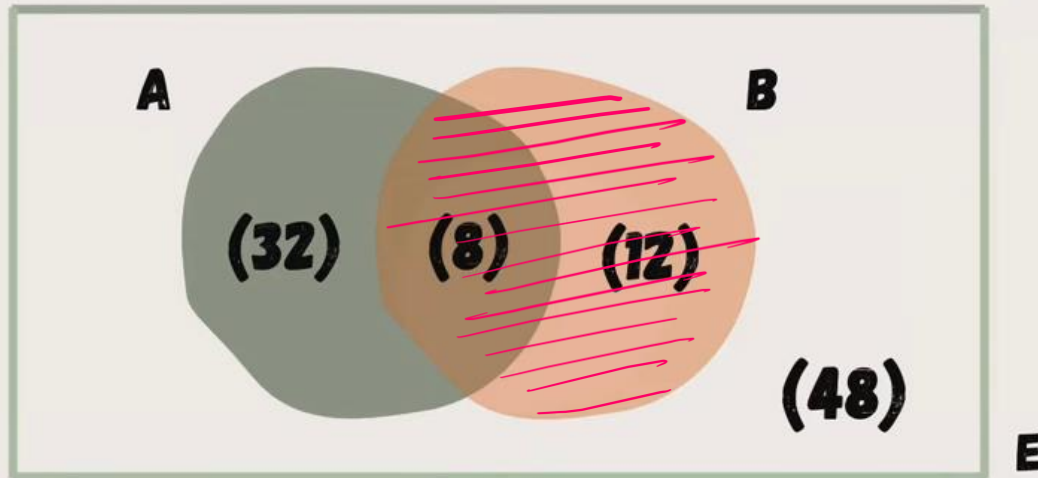
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$$P(A|B) = \frac{8}{20} = 0.4$$

Independent Events

INDEPENDENT EVENTS

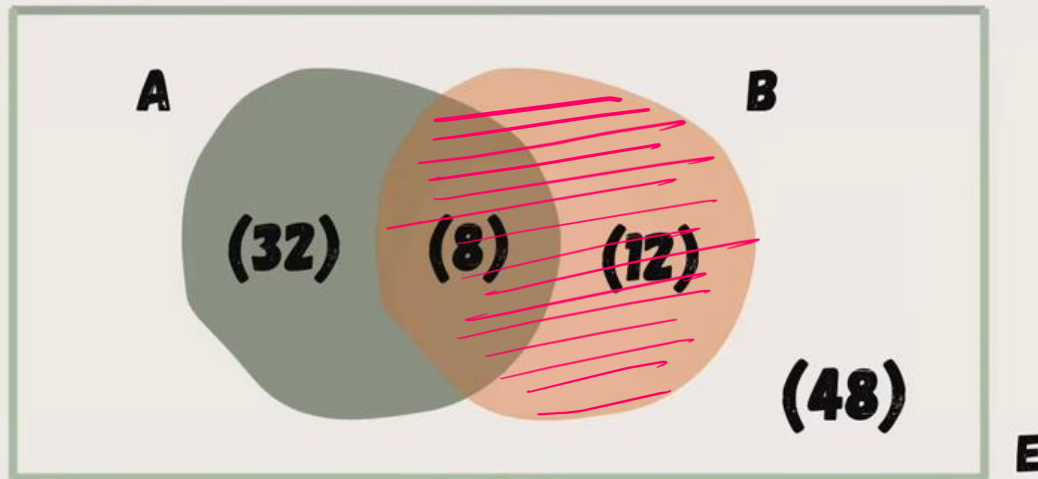
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$$= \frac{P(A \cap B)}{P(B)}$$

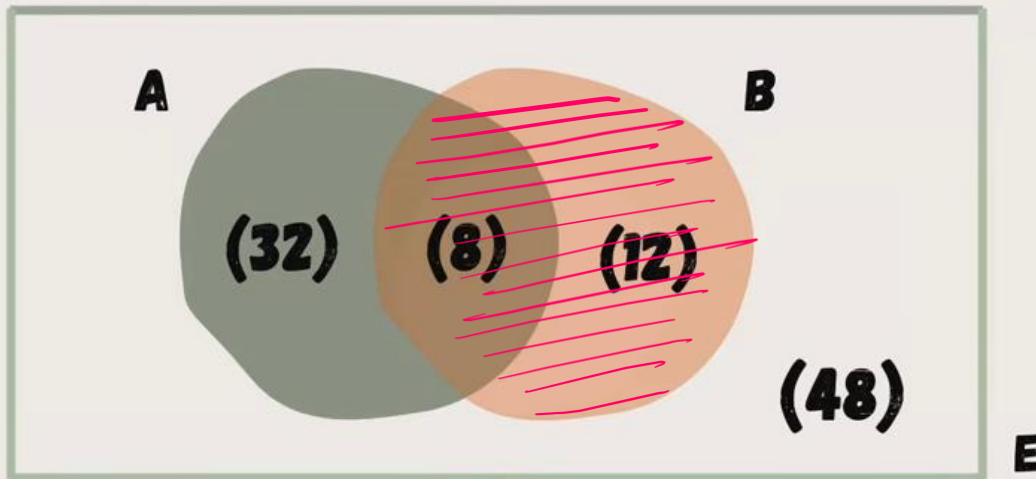
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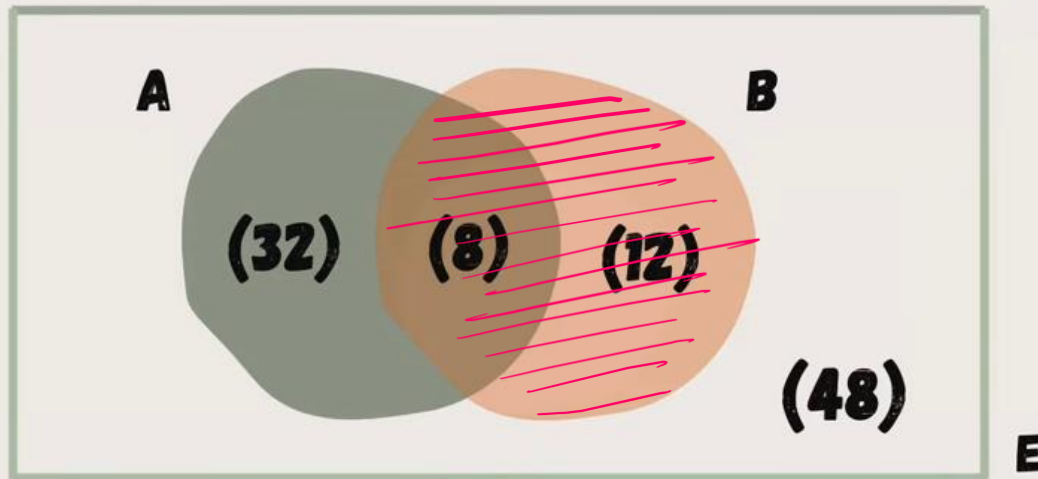
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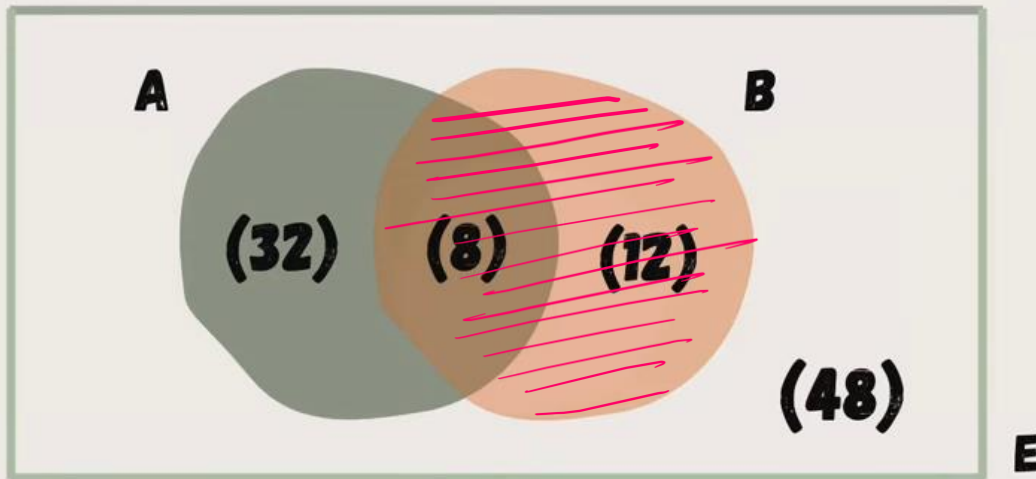
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$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

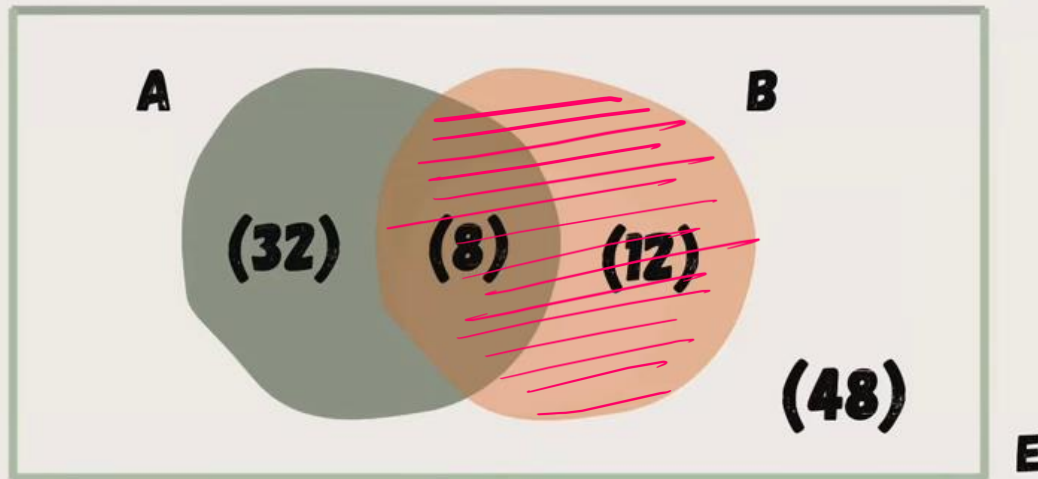
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same

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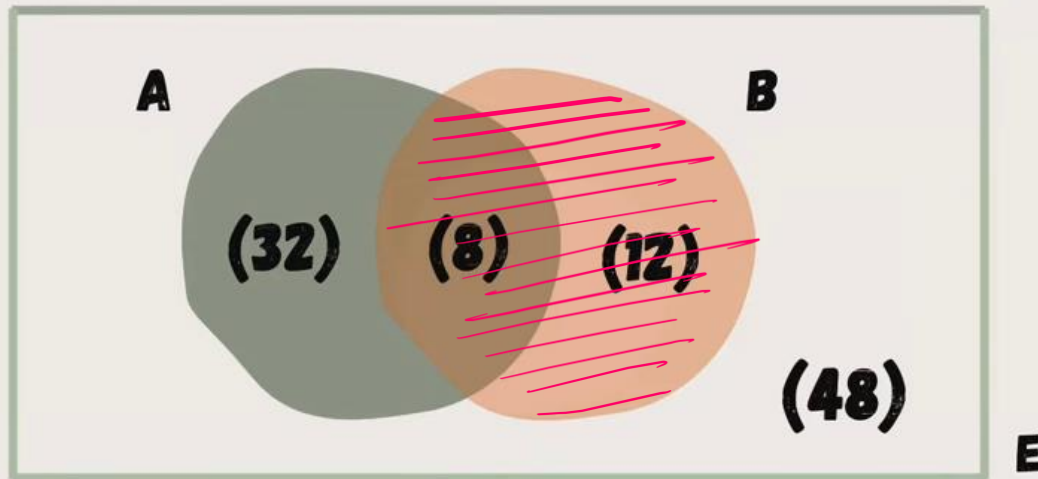
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same

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$$P(A/B)$$

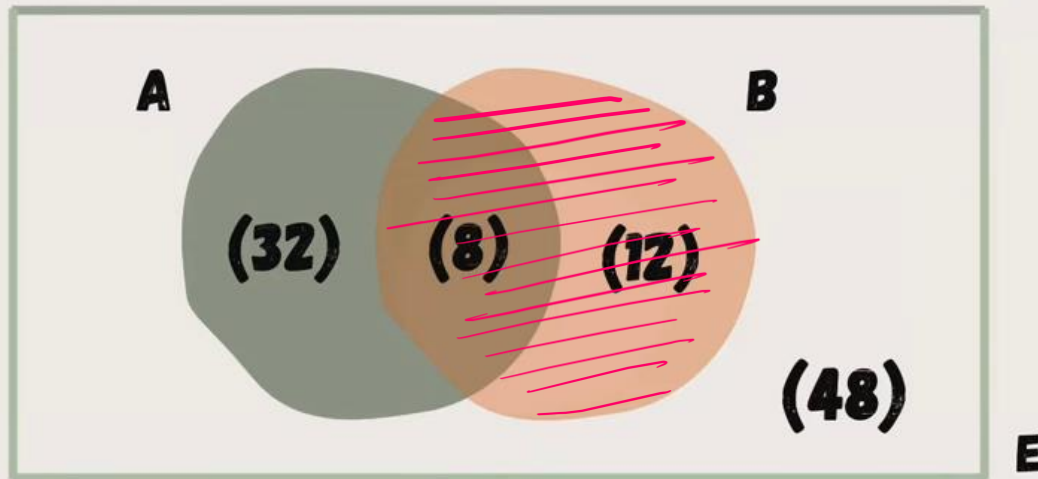
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$$P(A) = P(A/B)$$

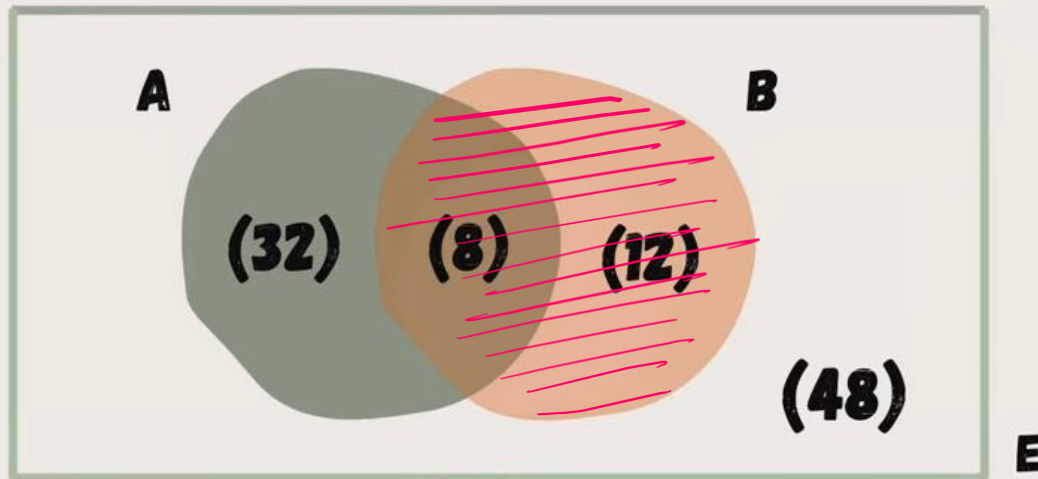
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same

$$= \frac{P(A \cap B)}{P(B)} = \frac{8/100}{20/100}$$

$$P(A) = P(A/B)$$

Similarly

$$P(B) = P(B/A)$$

Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student is taller
than 175 cm



Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student is taller
than 175 cm



$$P(A) = \frac{40}{100} = 0.4$$

Independent Events

Q: are these events independent?

INDEPENDENT EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
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EVENT B:
A student is taller
than 175 cm



$$P(A) = \frac{40}{100} = 0.4$$

$$P(A|B) = \frac{35}{50} = 0.7$$

Independent Events

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$$P(A) \neq P(A|B)$$

not independent

Independent Events

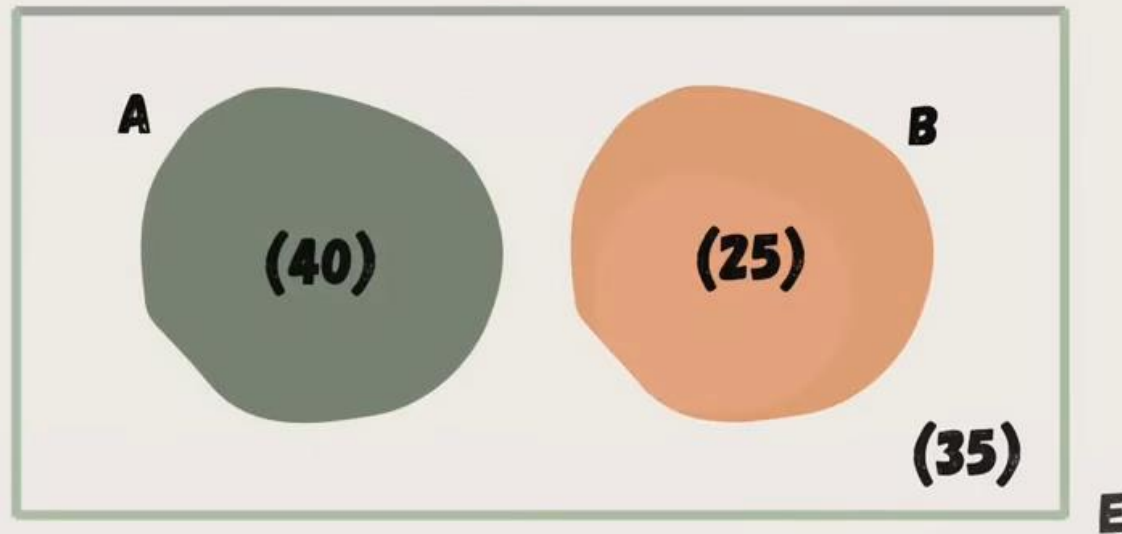
Q: are these events independent?

MUTUALLY EXCLUSIVE EVENTS

EXAMPLE: 100 highschool students are surveyed

EVENT A:
A student plays
Basketball

EVENT B:
A student plays
Cricket



Independent Events

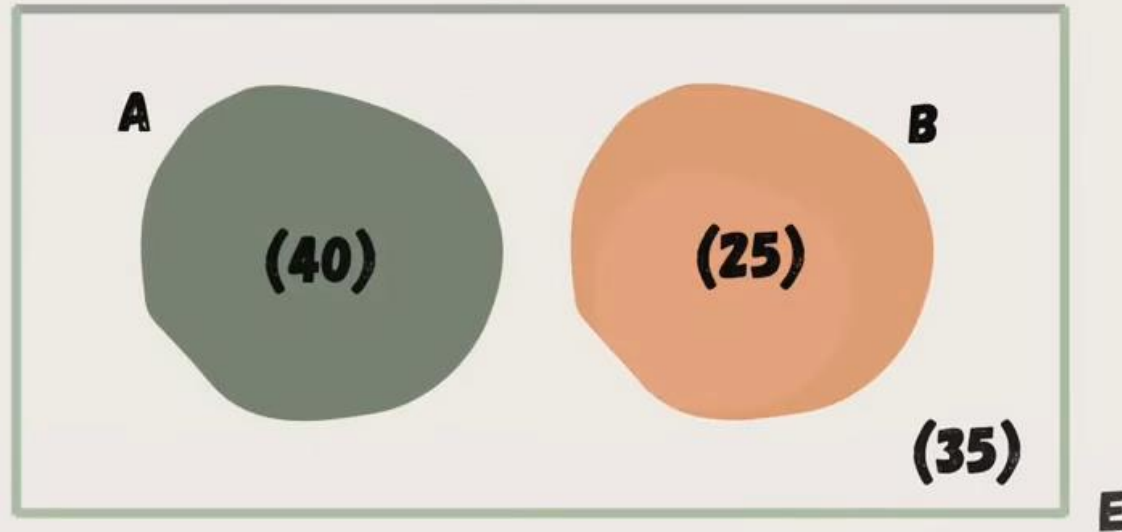
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Independent Events

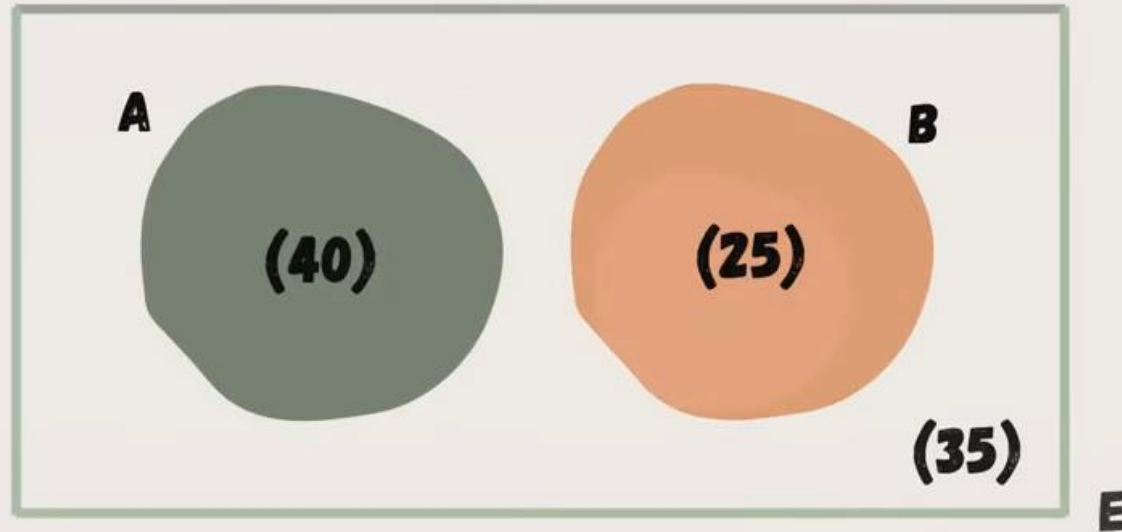
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$$P(A/B) = 0$$

Independent Events

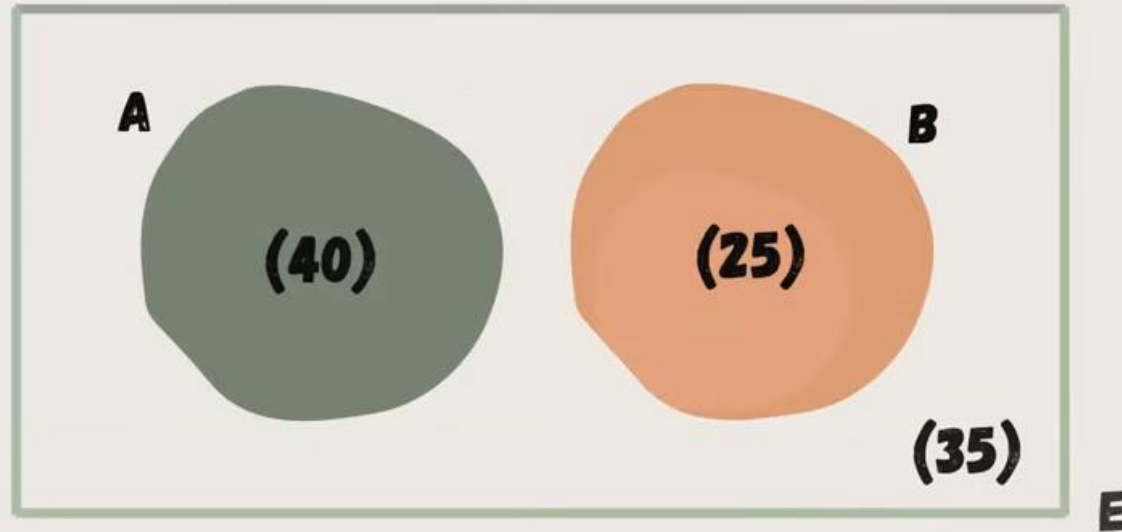
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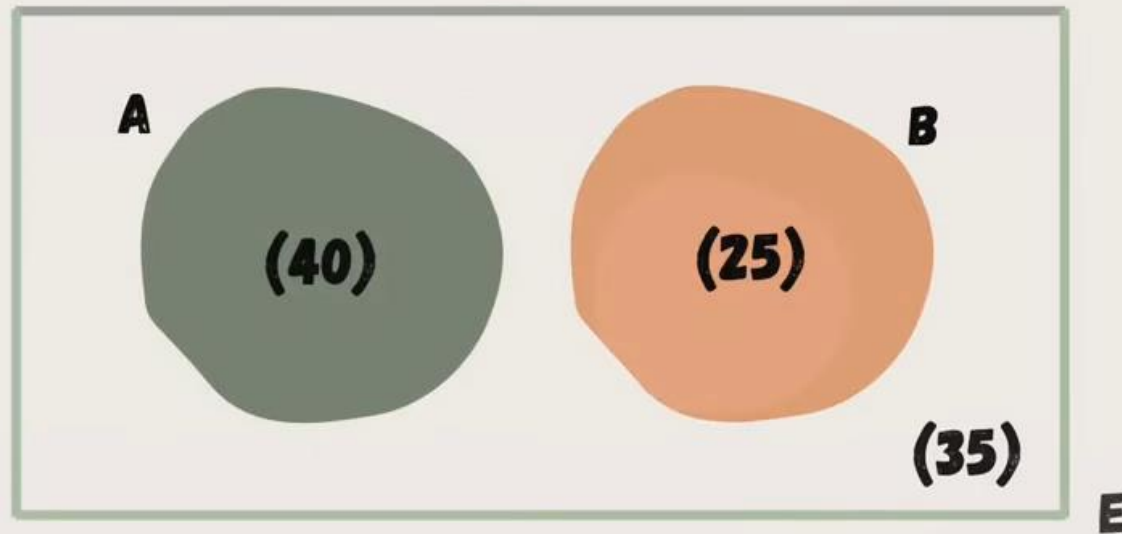
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not independent



Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?



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